CodeNeuraX

Bharatiya Antariksh Hackathon 2025

Problem Statement: Simulation/Modelling of Forest Fire Spread using AI/ML techniques
Description: Uncontrolled forest fires represent a significant challenge for government agencies tasked with preserving biodiversity and maintaining air quality. The spread of such fires is influenced by factors including weather conditions (temperature, precipitation, humidity, wind), terrain (slope, aspect, fuel availability), and human activity. With modern geospatial technologies, datasets from the Forest Survey of India and global services like *VIIRS-SNP* are accessible. Despite this, real-time simulation and forecasting remain complex. Short-term forecasting and dynamic simulation are crucial for timely preventive measures. AI/ML techniques offer promising capabilities to extract insights, helping planners estimate damage, prioritize containment, and mitigate fire impacts.

Problem Statement: AI-based Help Bot for Information Retrieval from a Knowledge Graph Based on Static/Dynamic Web Portal Content
Description: The MOSDAC (www.mosdac.gov.in) portal hosts satellite data and services accessible to the public. It includes extensive FAQs, documentation, product specifications, and support material. However, users often struggle to locate precise information due to layered navigation, mixed content formats, and time constraints. An AI-based help bot is envisioned to intelligently parse all hosted content and return instant, contextual responses to user queries, enhancing usability and knowledge accessibility across the platform.

Problem Statement: Monitoring Air Pollution from Space using Satellite Observations, Ground-Based Measurements, Reanalysis Data, and AI/ML Techniques
Description: Air pollution is a major environmental concern in many urban centers. Regular, consistent monitoring of air pollutant concentrations remains a challenge. Satellite remote sensing enables observations over broad spatial areas, allowing indirect estimation of pollutant levels. This problem statement focuses on

estimating surface-level particulate matter (PM) concentrations by leveraging satellite-derived data and AI/ML-based predictive modeling approaches.

Problem Statement: Designing a Chain-of-Thought-Based LLM System for Solving Complex Spatial Analysis Tasks Through Intelligent Geoprocessing Orchestration

Description: Geospatial challenges such as flood risk assessment, site suitability analysis, or land cover monitoring often demand sophisticated workflows involving various GIS tools and data sources. Traditionally, constructing these workflows requires expert knowledge and manual processes. This challenge invites participants to build a system that uses reasoning capabilities of Large Language Models (LLMs) to automatically plan and execute geospatial workflows—step-by-step—much like a human expert.

Problem Statement: Chase the Cloud: Leveraging Diffusion Models for Cloud Motion Prediction using INSAT-3DR/3DS Imagery

Description: This project focuses on predicting cloud motion using diffusion models, a class of deep generative networks, applied to satellite imagery from INSAT-3DR/3DS. Traditional optical flow or physics-based models often falter when confronted with volatile weather dynamics. In contrast, this approach uses spatio-temporal learning to simulate realistic cloud evolution from multi-spectral past frames, delivering enhanced forecasting for short-term (0–3 hours) applications including *nowcasting* and early warnings for severe weather events.

Problem Statement: AI/ML-Driven Automated Feature Detection and Change Analysis of Glacial Lakes, Road Networks, and Urban Drainage Systems from Multi-source Satellite Imagery

Description: This challenge aims to automate the detection, delineation, and spatiotemporal analysis of critical geospatial features—specifically glacial lakes, national highway centrelines, and urban drainage networks—by leveraging AI/ML/DL techniques. By integrating optical, SAR, and elevation data, participants are expected to build robust, scalable pipelines capable of generalizing across diverse landscapes and temporal variations. The resulting outputs will support infrastructure planning, environmental monitoring, and disaster risk reduction initiatives.

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Problem Statement: Developing an Algorithm for Air Quality Visualizer and Forecast App to Generate Granular, Real-time, and Predictive Air Quality Information
Description: An intuitive air quality visualizer and forecasting tool designed to bridge the data gap for underserved regions. While most AQI (Air Quality Index) apps focus on large metropolitan areas, this idea aims to develop an application that delivers hyperlocal, real-time, and predictive air quality insights—even in small towns and rural areas.

Problem Statement: Novel Approaches for Optimizing Deep Learning in Earth Observation with Imbalanced Data
Description: In many remote-sensing segmentation tasks, weaker classes (e.g., small water bodies, wetlands, deforested patches) are vastly underrepresented compared to dominant classes (e.g., vegetation, urban). Standard cross-entropy with SGD/Adam drives the network toward majority labels, yielding poor recall on rare targets. In this hackathon, participants will devise mathematically motivated optimizers, loss functions, or network modifications that explicitly address class imbalance in satellite imagery segmentation. The goal is to maximize weaker-class detection (e.g., IoU/F1) while preserving overall accuracy.

Problem Statement: Developing an AI/ML-based Algorithm for Identifying Tropical Cloud Clusters Using Half-Hourly Satellite Data from the INSAT
Description: Tropical Cloud Cluster (TCC) is a key element of the weather and climate by transporting mass, momentum and heat vertically into the Earth's atmosphere. Identifying the TCC is paramount important to the identification of cyclogenesis and its evolution is of great significance in weather and climate system. To study the initiation and development of TCCs over the Indian Oceanic region, a dataset called "Indian Tropical Cloud Cluster (ITCC)" to be generated using INSAT-3D Infrared Brightness Temperature (IRBRT) data. In order to generate the ITCC data, an automatic algorithm to be developed based on deep learning/machine learning. Thus, the current proposal describes methodology for identifying the TCCs using INSAT-3D IRBRT data. Identified TCCs will be tracked as they move around their Indian Ocean basin and variables such as TCC

size, location, convective intensity, cloud-top height. The algorithm can be adapted to near-real-time tracking of TCCs, which could be of great benefit to the tropical cyclone forecast community.

Problem Statement: Identifying Halo CME Events Based on Particle Data from SWIS-ASPEX Payload onboard Aditya-L1
Description: Our Sun drives the particle flux in the interplanetary medium and also affects our near earth environment. A sudden change in these particle flux can result in significant disturbance in the upper atmosphere and can result in major loss to the space assets. An early warning system based on particle measurement carried out at location much farther away, before these particles reach earth, can help us to take precautionary measures to prevent any catastrophic destruction.

Problem Statement: Novel method to detect landslides & boulders on the Moon: Using Chandrayaan images
Description: Landslides are mass movement rocks, debris of material downwards along the slope under the influence of gravity triggered by multiple geological activities. On Moon, there are plenty of surface with steep slope which are vulnerable to landslides and boulder falls. The identification of landslides and boulder fall are crucial because they indicate information about the recent activities undergone. Through this project, develop a novel algorithm using the Chandrayaan 1, 2 images to detect the landslide and boulder falls. The landslide occurs in wide range of size and extent and their pattern of occurrence is unpredictable. The Chandrayaan images provides a wide range (latitude, longitude) of coverage and using those images, the landslide and boulder fall presence need to be detected. The problem statement mentions a “Novel” algorithm, to detect the landslides means: the algorithm should show some novelty in landslide detection from the conventional or previous existing landslide detection techniques. The algorithm should also detect individual boulders in the given image and provide statistics like diameter or length and other statistics. After detection, the algorithm should be capable to decode the source for the landslide and provide vital information about it. The most challenging task is to look for signature and pattern in landslide and boulder fall and use it to highlight the most active region. Overall, the algorithm

should provide new information about the landslide & boulder fall once it is detected.

Problem Statement: Dual Image Super Resolution for High-Resolution Optical Satellite Imagery and its Blind Evaluation
Description: Acquiring high-resolution data from satellite platforms is an expensive operation due to resource constraints. Hence, satellites acquire multiple low-resolution images often shifted by half a pixel in both along and across track directions. These low-resolution images are utilized to generate high-resolution images. The quality of high-resolution images often depends on various satellite system parameters including the algorithm used for super-resolving. However, assessing the quality of these generated super-resolved images is challenging due to the absence of ground-truth references. This also necessitates the use of blind (no-reference) quality assessment techniques that can evaluate both the perceptual realism and fidelity of super-resolved images.

Problem Statement: Generation of High-resolution Lunar Digital Elevation Model from Lunar Images using Photoclinometry (Shape from Shading)
Description: Photoclinometry (also known as Shape-from-Shading, or SPC) is a technique used to extract topographic information from images acquired by spacecraft. 3D reconstruction of planetary surfaces using mono images, with appropriate illumination and viewing direction metadata, is essential for generating high-resolution DEMs, particularly where stereo imagery is unavailable. This technique not only enables DEM generation but also improves the accuracy of existing elevation datasets.

Problem Statement: Robust Change Detection, Monitoring, and Alert System on User-Defined AOI Using Multi-Temporal Satellite Imagery
Description: Change detection using satellite imagery plays a critical role in monitoring anthropogenic and environmental changes. Despite its potential, operational implementation faces bottlenecks such as cloud occlusions, atmospheric noise, and the difficulty in separating natural vs. human-induced changes.Imagery (~5m resolution with Red, Green, and NIR bands) from Bhoonidhi serves as an ideal input for high-frequency change analytics. The

objective is to build a robust, scalable platform capable of proactive monitoring and alerting for user-defined Areas of Interest (AOIs), filtering out irrelevant seasonal or natural fluctuations.